

Measurement failures and prospective safeguards in the empirical case

Observed failure	Scientific threat	Prospective safeguard	Empirical anchor
1. Parent-value leakage	Designated context not computationally necessary	Remove parent value from current prompt; verify prompt-alone closure	Earlier R2/MCAL S/T prompts reprinted parent values
2. Scorer censoring	Completed/readable model errors could leave denominator	Exact deterministic scorer; retain all completed/readable nonexact outputs as errors	Prospective scorer repair after scorer audit
3. Stochastic finalizer	Second model call could change worker measurement	Remove unnecessary stochastic finalizer	Finalizer removed prospectively
4. False all-zero derived ledger	Extraction erased completed raw behavior	Persist raw custody; reconstruct deterministically with frozen oracle/scorer/analyzer	336 raw responses preserved; corrected 138/168 LOCAL vs 131/168 CROSS; p=.29398
5. 2048 resource truncation	Planned response absent because output cap exhausted	Classify integrity failure and stop; no replay/resume of exposed bank	Ordinal 335 incomplete max_output_tokens; no confirmatory prefix
6. 8192 prospective requalification	Resource change after exposure could become outcome-driven tuning	Fresh LOCAL-only non-confirmatory qualification before new bank	32/32 complete/readable; zero integrity failures
7. Fresh fixed-bank confirmation	Outcome-informed exploratory signal could be mistaken for confirmation	Wholly fresh N=124 bank; frozen L=(D_U-D_S)/2; no retry/replacement	1,488/1,488 complete; depth primary +0.016129, p=.5368; escalation stopped

Historical states were preserved; the sequence represents measurement hardening, not accumulation of positive evidence.